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Capstone Project

7-step Problem Solving Model

1. Problem Identification (ID)
2. Reframe
3. Brainstorm
4. Choose 3
5. Advantages & Disadvantages
6. Prepare and Recommend
7. Engage Feedback



Problem

- Any significant, perplexing and challenging situation, real or artificial, the solution of which requires reflective thinking.
- A perplexing situation after it has been translated into a question or series of questions that help the direction of subsequent inquiry.



Step 1: Problem ID

Questions to Ask:

1. Who?
2. What?
3. When?
4. Where?
5. Why?
6. How?



Who is Involved?

- IT administrators
- System developers
- University staff and faculty
- Students
- Vendors/Cloud service providers
- Business owners/managers
- System developers or software engineers
- Warehouse or store staff
- Suppliers
- Customers



What is the Problem?

- Users experience frequent downtime when accessing the university's online portal.
- Small businesses struggle to manage inventory efficiently, leading to stockouts or excess stock.
- Students find it difficult to track multiple assignments and deadlines across different learning platforms.
- A local government unit lacks real-time data for monitoring public service complaints.
- Many companies face cyberattacks due to weak password policies and authentication systems.



When is the Problem?

Frequent downtime in the university's online portal

When:

- During peak usage times (e.g., enrollment periods, exam weeks, or submission deadlines).
- When the system undergoes maintenance or experiences technical issues.
- During periods of unexpected high traffic (e.g., emergencies or announcements).

Inventory management challenges for small businesses

When:

- During peak sales seasons (e.g., holidays or promotions), when demand fluctuates.
- When there are disruptions in the supply chain.
- At times when manual tracking methods are used or systems are outdated.
- When restocking schedules are not aligned with sales trends.



Where is it Happening?

Frequent downtime in the university's online portal

Where:

- University's online systems or server infrastructure (e.g., web servers, cloud platforms).
- Campus IT environment that hosts the portal.
- Students' and faculty devices (e.g., laptops, mobile phones) when accessing the portal remotely or on campus.
- Network environments where internet service disruptions may impact access.

Inventory management challenges for small businesses

Where:

- Small business warehouses or storage facilities where stock is kept.
- Retail stores or physical shops where products are sold.
- Point-of-sale (POS) systems used to track inventory in real time.
- Supplier locations where restocking and deliveries are coordinated.
- Business offices where management monitors inventory reports.



Why is it Happening?

Frequent downtime in the university's online portal

Why:

- Overloaded servers during peak times (e.g., enrollment periods).
- Inadequate system capacity or outdated infrastructure.
- Poor load balancing between servers or cloud services.
- Lack of redundancy or fault-tolerance mechanisms.
- Scheduled or unscheduled maintenance causing disruptions.
- Cyberattacks (e.g., DDoS attacks) targeting the portal.

Inventory management challenges for small businesses

Why:

- Manual inventory tracking leading to human errors.
- Poor forecasting of supply and demand trends.
- Inefficient restocking processes or delays from suppliers.
- Limited use of technology or lack of an automated system.
- Storage issues causing product spoilage or mismanagement.
- Supply chain disruptions affecting inventory levels.



How is it Happening?

Frequent downtime in the university's online portal

How:

- High traffic overwhelms the server during peak usage, causing crashes or timeouts.
- Single points of failure (e.g., a central server) make the system vulnerable to disruptions.
- Maintenance tasks or system upgrades take the portal offline temporarily.
- Network disruptions interrupt the connection between users and the server.
- Insufficient load balancing prevents traffic from being distributed across multiple servers.
- Cyberattacks (e.g., Distributed Denial of Service - DDoS) flood the portal with malicious requests.

Inventory management challenges for small businesses

How:

- Manual tracking with spreadsheets leads to errors and discrepancies.
- Unreliable stock counts occur due to lack of real-time updates.
- Miscommunication with suppliers causes delays in restocking.
- Over-ordering or under-ordering results from inaccurate sales forecasts.
- Lack of visibility into current inventory status leads to stockouts or excess stock.
- Products are misplaced in the warehouse, creating further inconsistencies in stock levels.



Characteristics of a Research Problem

- Specific – The problem should be specifically stated.
- Measurable – It is easy to measure by using research instruments, apparatus, or equipment.
- Achievable – Solutions to a research problem are achievable or feasible.
- Realistic – Real results are attained because they are gathered scientifically and not manipulated or maneuvered.
- Time-bound – Time frame is required in every activity because the shorter the completion of the activity, the better.



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Reframe

Problem: Users experience frequent downtime when accessing the university's online portal.

Goal: Ensure 99.9% uptime for the university's online portal by developing a fault-tolerant system.

Problem: Small businesses struggle to manage inventory efficiently, leading to stockouts or excess stock.

Goal: Minimize stockouts and optimize inventory levels for small businesses through an efficient inventory management system.



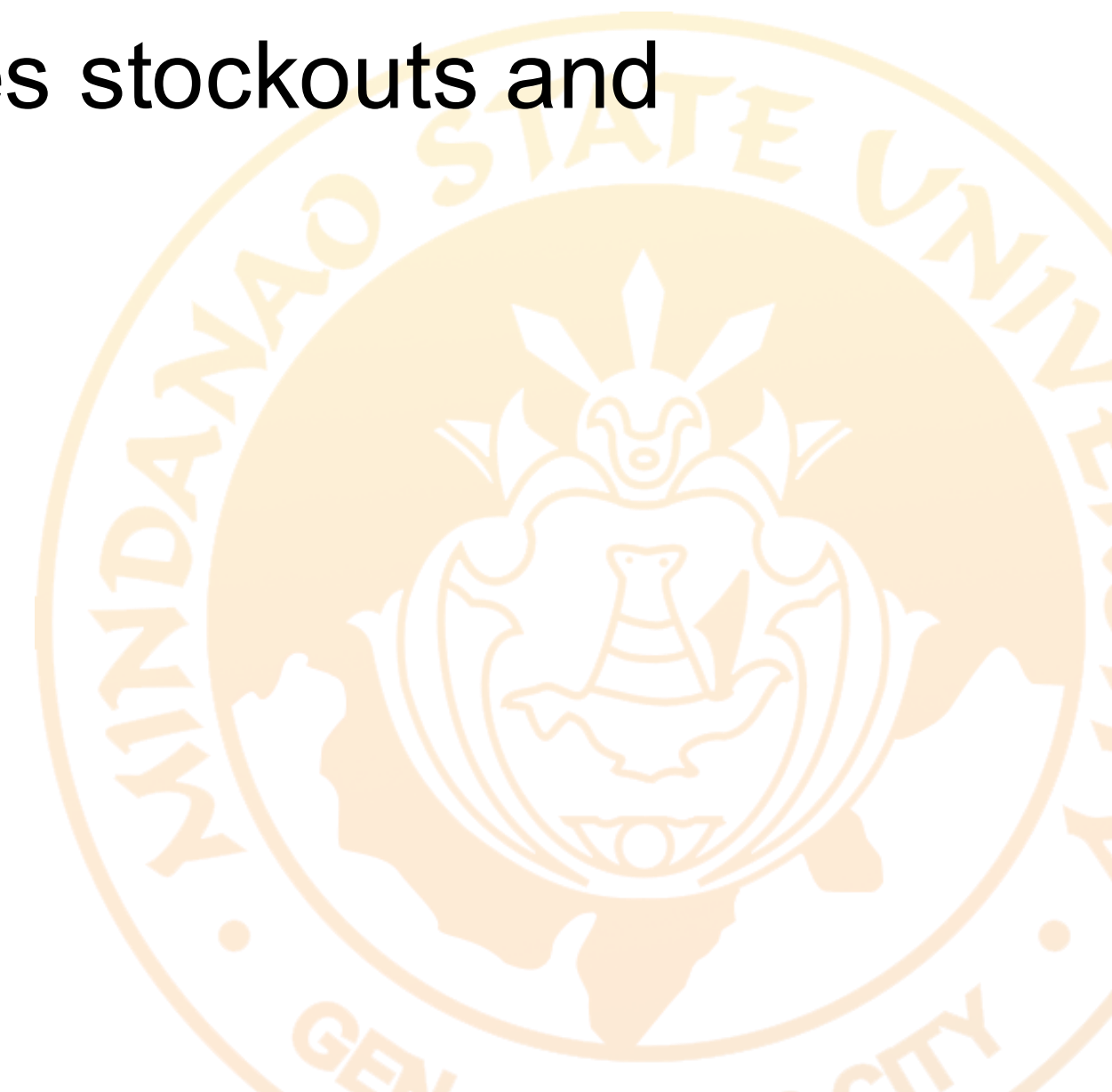
Reframe Goal into a Question

Problem: Users experience frequent downtime when accessing the university's online portal.

Goal: How can we develop a fault-tolerant system to ensure 99.9% uptime for the university portal?

Problem: Small businesses struggle to manage inventory efficiently, leading to stockouts or excess stock.

Goal: How can we design an inventory management system that minimizes stockouts and optimizes inventory levels for small businesses?



Criteria of a Good Research Problem

- Interesting
- Innovative
- Cost-effective
- Relevant to the needs and problems of the people
- Relevant to the government's thrusts
- Measurable and time-bound



Chapter 1

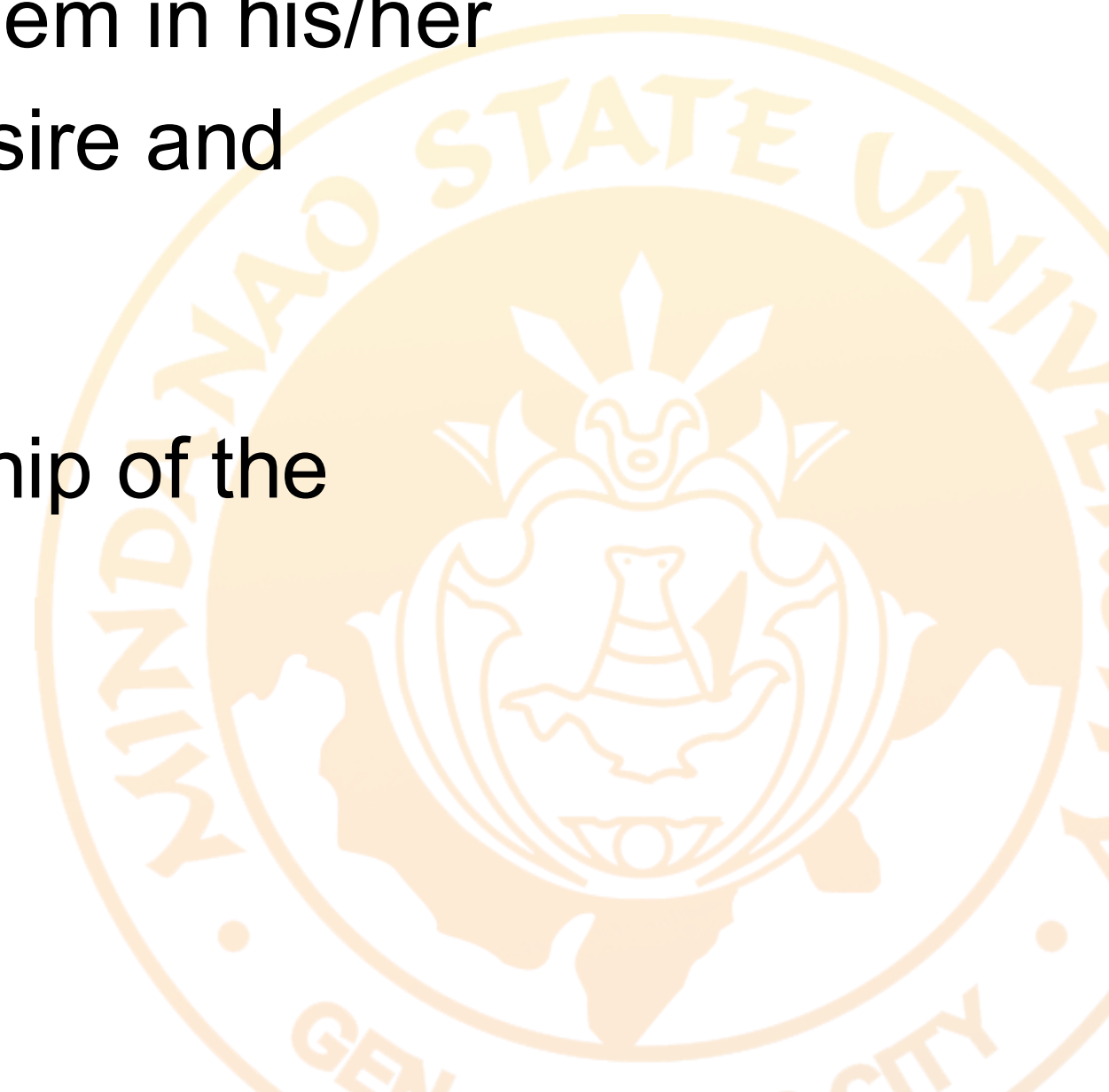
- Overview/Introduction/ Project context
- Purpose and description of the project
- Objectives of the Project
- Scope and Limitations of the Project
- Significance of the study/project



Overview/Introduction/Project Context

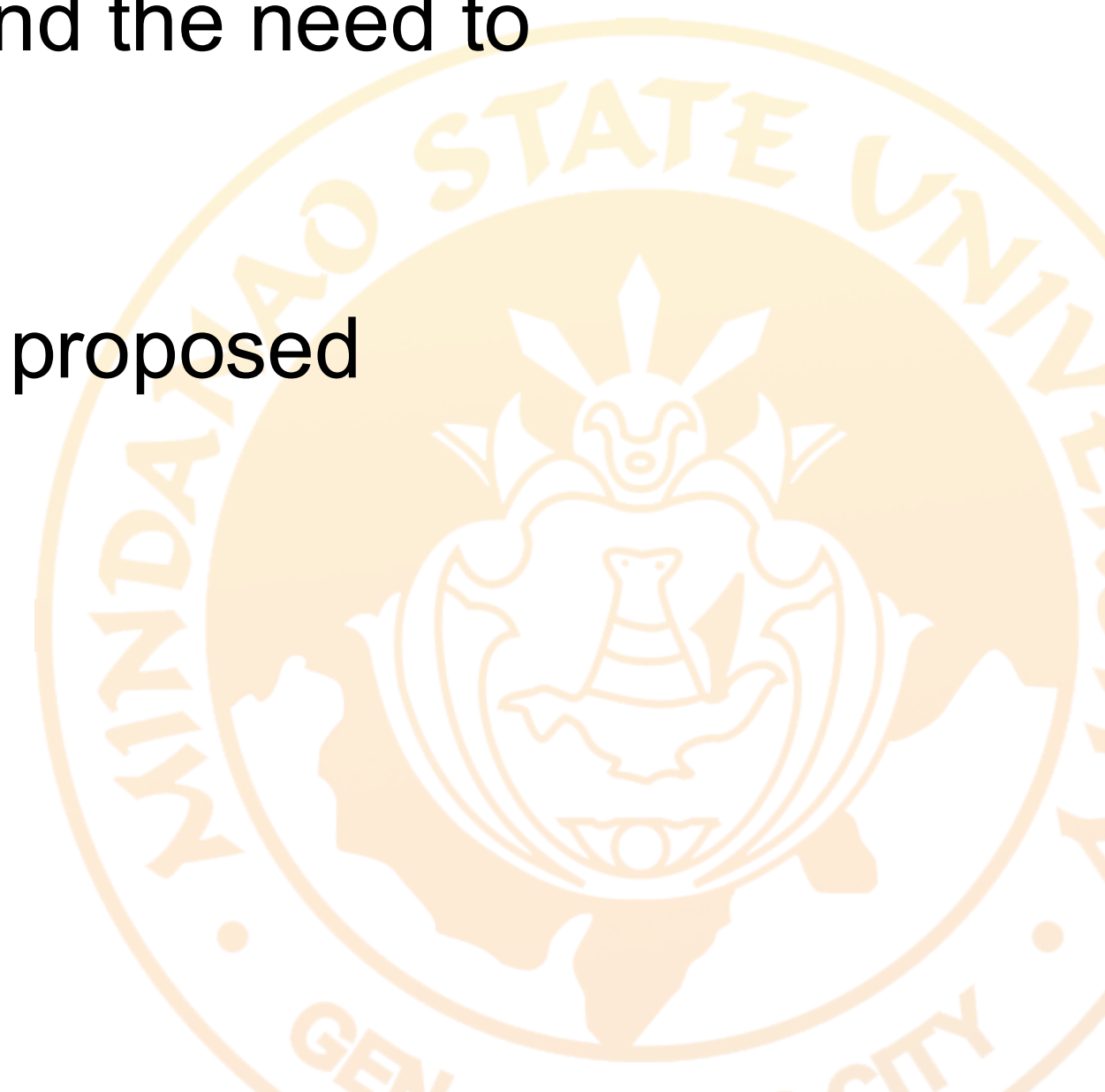
(must be at least 2-3 pages of presentation and discussion of...)

- The proponent should introduce the presentation of the problem, that is, what is the problem is all about. The proponent should describe the existing and prevailing problem situation based on his or her experience or from other literature. This scope may be global, national, regional and local.
- The proponent should give strong justification for selecting such research problem in his/her capacity as a researcher. Being part of the organization or systems and the desire and concern to improve the systems.
- The researcher state a sentence or two that would show the link and relationship of the rationale of the study to the proposed research problem.



Overview/Introduction/Project Context

- Presents the general context of the research supported by an international point of view or literature. It should include your motivation, synthesis of the literature, every paragraph should have its own, should not be repeated.
- The discussion must show literatures about the proposed system or software where innovation is the center of the presentation.
- Finally, end it with a clincher by justifying why you need to conduct this study and the need to develop a system emphasize the thesis statement of your study.
- Project context – this part of the document presents the current situation of the proposed project also known as the rationale.
 - Detailed discussion of the processes.
 - Show evidence – Interview, survey, field notes from the observation.



Overview/Introduction/Project Context

- Framework of the study – there are two major components of a framework of the study; these are the theoretical framework and conceptual framework.
- A theoretical framework is a structure that holds or supports a theory of a study. It is where you will describe the theory which explains why the research problem under study exists.
- A conceptual framework shows the graphical representations of the relationships and the processes of how the problems and the theories applied in the study.



Purpose and Description of the Project

- What is the function of the project?
- What is good in your project?
- What makes your project unique, unique, innovative, and relevant?



What makes a GOOD research proposal

- Relevance, either to the work of the funding body or to the student's course.
- The research is unique, or offers new insight or development.
- The title, aims and objectives, or offers new insight or development.
- Comprehensive and thorough background research and literature review has been undertaken.
- There is a good match between the issues to be addressed and the approach being adopted.



What makes a GOOD research proposal (Cont)

- The researcher demonstrates relevant background knowledge and/or experience.
- Timetable, resources and budget have all been worked out thoroughly, with eventualities covered.
- Useful policy and practice implications.



Reasons Why Research Proposals FAIL

- Aims and objectives are unclear or vague.
- There is a mismatch between the approach being adopted and the issues to be addressed.
- The overall plan is too ambitious and difficult to achieve in the timescale.
- The researcher does not seem to have conducted enough in-depth background research
- Problem is not that important.



Reasons Why Research Proposals FAIL (Cont)

- Information about the data collection is insufficiently detailed.
- Information about the data analysis method is insufficiently detailed.
- Timescale is inappropriate or unrealistic
- Resources and budget have not been carefully thought out.
- This topic has been done too many times before – indicates a lack in background research.



Thank you!

